K S V S Sobhita

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Summary

ECE undergraduate with interests in robotics, embedded systems, VLSI design, and intelligent hardware-software integration. Experienced in SystemVerilog-based hardware design and verification, FPGA development, and ESP32-based IoT systems. Currently developing an autonomous drone control framework integrating LLM/VLM reasoning, real-time control, and hardware-in-the-loop simulation. Seeking opportunities in robotics, embedded systems, RTL design, and intelligent hardware development.

Education

Amrita School of Engineering Bengaluru, India
B.Tech in Electronics and Communication Engineering *2023 – Present*
GPA: **7.55/10.0**

Sri Venkateswara Jr. College Visakhapatnam, India
Intermediate (MPC) *2021 – 2023*

Technical Skills

- C++
- C Programming
- CoppeliaSim
- Verilog
- Python
- Embedded Systems
- Linux
- MATLAB/Simulink

Experience

Elevance Skills Remote
Electronics & Robotics Intern *May 2026 – Oct 2026*

- Developed an ESP32-based IoT healthcare monitoring system integrating FreeRTOS, MQTT communication, and role-based dashboards for patient vitals and environmental monitoring.
- Developed smart healthcare control simulations for medication dosage adjustment and bed elevation, using queues, semaphores, and mutexes for inter-task communication and data consistency.

Projects

AERIS – Autonomous Drone Operation Using LLMs and VLMs *Aug 2026 – Present*
GitHub: AERIS_llm_vlm_drone

- Developing a closed-loop autonomous drone control system integrating LLM/VLM reasoning with a LiteWing ESP32 drone over Wi-Fi for natural-language command execution.
- Implementing vision-based spatial grounding to map 2D camera coordinates to 3D waypoints, with dynamic re-planning and trajectory safety validation.
- Building a telemetry platform using FastAPI, React, and WebSockets, alongside MATLAB/Simulink hardware-in-the-loop simulation.

Robotic Car Painting Cell – 6-DOF UR10 Simulation *Oct 2026 - Oct 2026*
GitHub: robotic_car_painting_cell

- Simulated an automated painting cell in CoppeliaSim with a 6-DOF UR10 via D-H kinematics.
- Engineered a Python serpentine toolpath planner maintaining 250 mm standoff and surface-normal alignment.
- Built an AABB collision detection module with line-segment sampling, ensuring 50 mm safety clearance.

Logic Analyzer Using FPGA *Jul 2025 – Apr 2026*
GitHub: Logic_Analyzer_using_FPGA

- Architected a modular FPGA-based logic analyzer in SystemVerilog on the Basys-3 board for real-time SPI, I2C, and GPIO protocol decoding.
- Designed configurable trigger logic for condition-based signal capture across multiple channels.
- Implemented a ring-buffer architecture for event storage and continuous signal monitoring.

Training & Certifications

Foundation – Physical Design *Open-Source EDA Tools*

- Studied physical design stages with hands-on labs in synthesis and design-for-test (DFT) using open-source tools.

Building Conversational AI Applications *NVIDIA Deep Learning Institute*

- Completed NVIDIA DLI "Get Ready" training, gaining foundational exposure to AI, deep learning, and GPU-accelerated workflows.